

SAI SREENITHI JESSU

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Summary

Driven Computer Science graduate with a strong foundation in software development, system architecture, algorithmic problem-solving, and mathematics, complemented by hands-on projects. Eager to learn and apply generative AI, LLMs, and predictive modeling to deliver tailored, analytics-based solutions and drive intelligent automation. Actively pursuing AI courses to deepen understanding of advanced concepts and practical applications.

Education

University of Waterloo

2021 – 2025

Bachelor of Science in Computer Science

Waterloo, ON

Relevant Coursework

- | | | | |
|--------------------------|----------------------|----------------------|------------------------|
| • Computer Organization | • Data Abstraction | • Combinatorics | • Business and Finance |
| • Functional Programming | • Operating Systems | • Compilers | • Macroeconomics |
| • Algorithm Design | • Object-Oriented SD | • User Interfaces | • Microeconomics |
| • Database Management | • Data Structures | • AI App Development | • Statistics |

Experience

BlackBerry

Jan 2023 – April 2023

Software Development and Cybersecurity Project Management Intern

Waterloo, ON

- Automated business workflows using C, Active Directory, and SQL Server, optimizing processing for 65,000+ annual requests and improving operational efficiency.
- Led a SOC2 compliance initiative, collaborating with cybersecurity leaders to enhance audit readiness and IT risk posture, achieving a 90% compliance rating.
- Coordinated penetration testing and vulnerability assessments, aligning with IT audit standards and incident response protocols to strengthen organizational security.
- Developed and maintained weekly security metrics and dashboards, providing actionable insights for the CISO and VP of IT; a graphic was later featured in a presentation to the CEO.
- Improved identity and access management for 100+ employees through PowerShell automation and Active Directory scripting, increasing system reliability and access control.
- Created executive-level documentation and presentations for the CIO and CISO, translating complex technical findings into clear insights for leadership.
- Collaborated with product managers and engineering leads to integrate security goals into the product roadmap and development cycle.
- Actively contributed to multiple teams and earned a spot in the business automation group for proactive security-focused process design.
- Recognized for exceptional contributions by the CISO and Executive VP of BlackBerry and received a formal farewell acknowledgment from the CIO at the time.

Projects

StudyBuddy – Concept Design for AI-Powered Study Matchmaking

May 2025 – August 2025

Project Manager, CS 449/649 Human-Computer Interaction Course Project

University of Waterloo

- Led a 4-person team in conceptualizing *StudyBuddy*, a mobile app designed to address academic disengagement and isolation by connecting students for real-time study sessions.
- Explored how AI-driven matchmaking algorithms could leverage course enrollment, availability, and study preferences to recommend compatible study partners.
- Applied HCI research methods including heuristic evaluations, cognitive walkthroughs, and user interviews to iteratively refine user flows and interface concepts.
- Created low-fidelity wireframes and interactive high-fidelity prototypes in Figma to simulate app functionality such as swipe-to-match and study-style filtering.
- Framed AI not as an implementation but as a problem-solving tool, showcasing ability to translate academic and social challenges into viable AI-enabled design solutions.

CastorOS OS Kernel | *C, Assembly, x86, Bash, QEMU*

May 2025 - August 2025

- Built and extended *CastorOS*, a teaching kernel in C and x86 Assembly, simulating a minimal operating system on QEMU.
- Implemented core OS components including process management ('spawn' system call), memory loading, page table handling, and system call interfacing with user-space binaries.
- Configured custom system call handlers and modified the OS loader to support dynamic process initialization, isolation, and context switching.
- Used SCons for kernel builds and debugging tools like 'kprintf', GDB, and QEMU tracing to test functionality and resolve segmentation issues.
- Gained hands-on experience in memory protection, privilege-level transitions, and kernel-user separation on an emulated x86 system.

SereneDesk: AI-Backed Workspace Mood Optimizer | *Python, OpenAI Whisper, GPT-4*

Feb 2025 – Present

- Building a Streamlit dashboard that monitors ambient noise and transcribes user voice check-ins to identify stress patterns and triggers.
- Applying GPT-4 sentiment analysis to generate contextual wellness suggestions, journaling prompts, and ambient soundscapes.
- Processing and analyzing sentiment data with Python and pandas to ensure clean, structured, and interpretable results.
- Implementing interactive dashboards with Plotly and Streamlit to visualize stress and wellness patterns.
- Applying structured problem-solving to translate stress and wellness challenges into technical solutions from data ingestion to AI-driven recommendations.
- Building monitoring and evaluation frameworks to track GPT-4 sentiment analysis accuracy and recommendation effectiveness.
- Collaborating with users to gather feedback and iteratively improve feature functionality and overall user experience.
- Designing scalable architectures to support multiple users and real-time insight generation.

WLP4 Compiler for MIPS Assembly | *C++, Assembly, Nano, Bash*

Jan 2025 - April 2025

- Designed and implemented a full compiler for the WLP4 language, compiling to MIPS assembly through a multi-phase pipeline consisting of scanning, parsing, semantic analysis, and code generation.
- Constructed a lexical analyzer using a DFA and Simplified Maximal Munch algorithm; built an SLR(1) parser from a grammar specification using LR tables.
- Integrated semantic type checking for pointer operations, arrays, and function call resolution; handled recursive descent translation to MIPS for control flow and memory operations.

PathFinder AI: Career Advisor for CS Students | *Python, Streamlit, GPT-4 API, JSON*

June 2024 – Present

- Developing an AI-powered career guidance tool that provides personalized suggestions to CS students exploring paths beyond software development.
- Implementing GPT-4 to generate tailored recommendations based on students' skills, interests, and career values.
- Integrating JSON-based persona profiling and dynamic conversation routing for context-aware guidance.
- Created interactive Streamlit prototypes to simulate AI responses and evaluate user interactions.
- Conducting user testing and gathering feedback to iteratively improve clarity and usability.
- Applied structured problem-solving to translate student career challenges into scalable technical solutions.

Chess Engine with Computer Opponent | *C++, VS Code, PowerShell, Nano*

Sep 2022 – Dec 2022

- Developed a full-featured chess engine in C++, applying object-oriented design principles (inheritance, polymorphism, encapsulation) to structure pieces, moves, and game rules.
- Implemented an AI opponent by coding move-generation, evaluation heuristics, and decision-making logic, enabling the computer to play competitively against human players.
- Integrated error handling, input validation, and modular design to ensure reliable gameplay and maintainable codebase.
- Strengthened debugging and testing skills by validating edge cases (check, checkmate, stalemate, castling, en passant) and refining AI move selection.

Technical Skills

Programming Languages: Python, Java, C, C++, Racket, Bash, C#, Assembly, R

Back-End & Scripting: Java (Spring MVC), C#, Python, PowerShell

Data & Analytics: SQL Server, PowerBI, pandas, data analysis, visualization, statistics

DevOps & Systems: Windows Terminal, Linux, Jira, Agile Methodology, CI/CD, SDLC

Tools & Productivity: Git, Eclipse, VS Code, Google Suite, MS Office (Excel, PowerPoint, Outlook)

Certifications

- Intro to AI in Azure - Microsoft Learn (Completed)
- Microsoft Certified: Azure AI Fundamentals (Completed)

Leadership

Founder & President, Telugu Students Association (TSA)

Sep 2023 – Present

University of Waterloo

Waterloo, ON

- Founded TSA when only four Telugu students were on campus, growing membership to 60+ active participants within one year.
- Launched dynamic social media campaigns and campus outreach initiatives, significantly increasing event visibility and attendance.
- Built inclusive networking and mentorship programs by pairing underclassmen with senior students for academic and career guidance.
- Directed strategic planning, delegation, and conflict resolution while managing an executive team with clear accountability.
- Positioned TSA as a bridge between cultural identity and community impact, transforming limited resources into a thriving, recognized organization.

Volunteer Organizer, Women in Computer Science (WiCS)

Jan 2022 – Present

University of Waterloo

Waterloo, ON

- Coordinated large-scale webinars and conferences with 100+ participants, delivering seamless technical support for guest speakers and attendees.
- Oversaw backend logistics including platform setup, breakout room management, live troubleshooting, and real-time session moderation.
- Expanded accessibility by ensuring session recordings, captions, and resources were distributed post-event.
- Designed and analyzed attendee feedback surveys to optimize future events, improving structure and attendee satisfaction.

Affiliations & Memberships

- Member, Computer Science Club, University of Waterloo
- Member, Product Manager Club, University of Waterloo
- Member, Stocks Club, University of Waterloo